

(3.2) a)

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$$u_1 = \frac{1}{\|1\|}$$

$$\sqrt{\int_1^5 1 dx} = \sqrt{4} = 2$$

$$u_1 = \frac{1}{2} = 0,5$$

$$w_2 = \frac{1}{2} \int_1^5 x \cdot \frac{1}{2} dx = \frac{1}{2} \left[\frac{x^2}{2} \right]_1^5 = 3 \quad \frac{x-3}{\|x-3\|} = w_2$$

$$\sqrt{\int_1^5 (x-3)^2 dx} = \sqrt{\frac{16}{3}}$$

$$u_2 = \frac{\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4}}{\sqrt{\frac{16}{3}}} = 0,433x - 1,299$$

$$w_3 = \frac{1}{2} \int_1^5 x^2 \cdot \frac{1}{2} dx + \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4} \right) \int_1^5 x^2 \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4} \right) dx = 6x - \frac{23}{3}$$

$$\|x^2 - w_3\|^2 = \int_1^5 \left(x^2 - 6x + \frac{23}{3} \right)^2 dx = \frac{256}{45} \quad \sqrt{\frac{256}{45}} = \frac{16}{3\sqrt{5}}$$

$$u_3 = \frac{x^2 - w_3}{\|x^2 - w_3\|} = \frac{3\sqrt{5}}{16}x^2 - \frac{9\sqrt{5}}{8}x + \frac{23\sqrt{5}}{16} = 0,419x^2 - 2,516x + 3,214$$

(u_1, u_2, u_3) tvoří ortonormální bázi V

$$\begin{aligned} b) \quad P &= \frac{1}{2} \int_1^5 \sin x \cdot \frac{1}{2} + \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4} \right) \int_1^5 \sin x \left(\frac{\sqrt{3}}{4}x - \frac{3\sqrt{3}}{4} \right) dx + \\ &+ \left(\frac{3\sqrt{5}}{16}x^2 - \frac{9\sqrt{5}}{8}x + \frac{23\sqrt{5}}{16} \right) \int_1^5 \sin x \left(\frac{3\sqrt{5}}{16}x^2 - \frac{9\sqrt{5}}{8}x + \frac{23\sqrt{5}}{16} \right) dx = \\ &\doteq \underbrace{0,064}_{1.} - \underbrace{0,647x + 1,940}_{2.} - \underbrace{0,053x^2 + 0,315x - 0,403}_{3.} = \end{aligned}$$

$$= -0,053x^2 - 0,332x + 1,601$$

num. aproximace více polynomů

c)

$$\| \sin x - (-0,053x^2 - 0,332x + 1,601) \| =$$

$$= \sqrt{\int_1^5 (\sin x - (-0,053x^2 - 0,332x + 1,601))^2} \doteq \underline{\underline{0,319}}$$